Uniaxial anisotropy on tetrahedral spin bodies: axis selection and exact effective $O(2)$ models

Note for the Polyhedron_Spin_Group follow-up

3 November 2026

1 Setup

We add a single-spin anisotropy to the $n = 4$ model of the main paper,

$$V = \sum_a \sum_{i=1}^{4} \big[a_2 P_2(\hat{z} \cdot \mathbf{S}_{a,i}) + a_4 P_4(\hat{z} \cdot \mathbf{S}_{a,i})\big], \tag{1}$$

with $a_2 = 2D/3$ for $D(S^z)^2$ and $a_4 = 8B/35$ for $B(S^z)^4$ (the P_2 part of $B(S^z)^4$ adds to a_2). V is a sum of single-body terms, so the uniform-state theorem still holds and the problem reduces to one body with pair potential $\Phi(x) = \sum_l \kappa_l P_l(x)$, $\kappa_l = f_l - z g_l$. The deformation stiffnesses λ_E, λ_{T_2} of Eq. (9) of the main paper therefore already contain the inter-body couplings. The symmetry is reduced from $O(3)$ to $D_\infty \times \mathbb{Z}_2^T$.

2 First order: the two-design removes a_2

Because the tetrahedron is a spherical 2-design, $\sum_i (\hat{z} \cdot \mathbf{S}_i)^2 = 4/3$ for every orientation, so the a_2 term has no first-order effect. The a_4 term is the $l = 4$ cubic harmonic. With $K \equiv \sum_\alpha n_\alpha^4$, where $\hat{n}$ is $\hat{z}$ in the cube frame of the body ($K = 1$ for an S_4 axis, $K = 1/3$ for a C_3 axis),

$$E^{(1)} = a_4 \Big(\frac{7}{3} - \frac{35}{9} K\Big). \tag{2}$$

3 Second order: selection through deformation

At first order the energy does not depend on orientation, so there is no torque. The force acts only on the E and T_2 deformation modes. The force on spin i is $3a_2(\hat{z} \cdot \mathbf{S}_i)\,[\hat{z} - (\hat{z} \cdot \mathbf{S}_i)\mathbf{S}_i]$. At the symmetric orientations it lies entirely in one representation:

$$S_4 \parallel \hat{z}: \quad |f_E|^2 = 4 \cdot 9 \cdot \tfrac{1}{3} \cdot \tfrac{2}{3}\, a_2^2 = 8a_2^2, \quad f_{T_2} = 0, \tag{3}$$

$$C_3 \parallel \hat{z}: \quad |f_{T_2}|^2 = 3 \cdot 9 \cdot \tfrac{1}{9} \cdot \tfrac{8}{9}\, a_2^2 = \tfrac{8}{3} a_2^2, \quad f_E = 0. \tag{4}$$

The 5-dimensional $l = 2$ force splits as $E \oplus T_2$ under T_d, with weights $(3K - 1)/2$ and $3(1 - K)/2$. The result is again linear in K:

$$E^{(2)} = -\tfrac{1}{2}\Big[\frac{|f_E|^2}{\lambda_E} + \frac{|f_{T_2}|^2}{\lambda_{T_2}}\Big] = -2a_2^2 \Big(\frac{3}{\lambda_E} - \frac{1}{\lambda_{T_2}}\Big) K + \text{const.} \tag{5}$$

In terms of derivatives of Φ at the tetrahedral angle, Eq. (9) of the main paper reads

$$\lambda_E = \tfrac{8}{3}\, \Phi''(-\tfrac{1}{3}), \qquad \lambda_{T_2} = \tfrac{8}{9}\big[3\Phi'(-\tfrac{1}{3}) + 2\Phi''(-\tfrac{1}{3})\big]. \tag{6}$$

Result.

$$\boxed{E_{\text{eff}}(\hat{n}) = -\Gamma \sum_{\alpha} n_{\alpha}^4, \qquad \Gamma = \frac{35}{9} a_4 + 2a_2^2 \Big(\frac{3}{\lambda_E} - \frac{1}{\lambda_{T_2}} \Big)} \tag{7}$$

$\Gamma > 0$ selects an S_4 axis along $\hat{z}$, $\Gamma < 0$ a C_3 axis. The second-order term does not depend on the sign of D (easy axis or easy plane). Direct numerical minimization for five random κ sets and three anisotropy combinations agrees with this formula to better than 1%.

4 Selection criterion: κ_2 cancels

$$3\lambda_{T_2} - \lambda_E = \frac{8}{27} (27\kappa_1 - 63\kappa_3 + 35\kappa_4) = \frac{8}{27} (27J + 9zG - 8zH_4). \tag{8}$$

- In the whole T phase of the (zG, zH_4) plane at $K = 0.5$ this is positive, so the deformation channel always selects $S_4 \parallel \hat{z}$.
- $C_3 \parallel \hat{z}$ requires either $a_4 < -\frac{18}{35} a_2^2 (3/\lambda_E - 1/\lambda_{T_2})$ (quartic single-ion anisotropy wins) or, at large K, the window $8zH_4 > 27J + 9zG$.
- As $K \to 0$, $\lambda_E \propto K \to 0$: perturbation theory fails and S_4 selection becomes strong.

5 Exact effective models of the selected phases

For a rigid tetrahedron only the $l = 3$ and $l = 4$ channels couple bodies ($l = 1, 2$ vanish for a 2-design), with $g_3 = \frac{2}{5} G$, $g_4 = \frac{8}{35} H_4$. The bond energy is $-g_l \frac{4\pi}{2l+1} \operatorname{Re} \mathbf{q}_a^{(l)} \cdot \mathbf{q}_b^{(l)*}$. The multipole weights (checked numerically) are:

axis	$l = 3$	$l = 4$
S_4	$m = \pm 2$: 80/9	$m = 0$: 196/81; $m = \pm 4$: 140/81
C_3	$m = 0$: 400/81; $m = \pm 3$: 320/81	$m = 0$: 784/729; $m = \pm 3$: 2240/729

$S_4 \parallel \hat{z}$: generalized XY model ($q = 2$)

With $\theta = 2\alpha$ (α the in-plane body angle),

$$H = -\sum_{\langle ab \rangle} \Big[\frac{32}{9} G \cos(\theta_a - \theta_b) + \frac{32}{81} H_4 \cos 2(\theta_a - \theta_b) \Big]. \tag{9}$$

Time reversal acts as $\theta \to \theta + \pi$. The polar phase is octupolar order; the half-vortex "nematic" phase is the time-reversal-even hexadecapolar phase, the 2D version of the $T_d \subset O_h$ (cubatic) vestigial phase. The mixing ratio is fixed microscopically, $\Delta/(1 - \Delta) = 9G/H_4$, so the vestigial phase needs H_4/G of order 20.

$C_3 \parallel \hat{z}$: coupled Ising–XY model

Let $\sigma = \pm 1$ label up/down tetrahedra (related by time reversal) and $\varphi = 3\alpha + \pi(1 - \sigma)/2$. Then

$$H = -\sum_{\langle ab \rangle} \Big[\frac{160}{81} G \, \sigma_a \sigma_b + \Big(\frac{128}{81} G + \frac{512}{729} H_4 \, \sigma_a \sigma_b \Big) \cos(\varphi_a - \varphi_b) \Big]. \tag{10}$$

- For $H_4 = 0$ the Ising and XY variables decouple exactly: the octupolar channel alone cannot mix $\sigma\sigma'$ with $\cos(\varphi - \varphi')$. Square-lattice estimates: $T_{\rm Ising} \approx 1.12\, zG$ above $T_{\rm KT} \approx 0.35\, zG$.
- H_4 switches on the coupling, $J_4/J_2 = \frac{4}{9}\, zH_4/zG$. At $J_4 = J_2$ the model has the Granato–Kosterlitz–Lee–Nightingale form studied for the fully frustrated XY model. The coupling is thus tuned continuously from zero by a single microscopic ratio.
- A 2π vortex of φ is a $2\pi/3$ vortex of the body; an Ising domain wall twists α by $\pi/3$.

Small Γ. In 2D the $SO(3)/T$ correlation length grows as $\exp(c/T)$, so the effective anisotropy $\xi^2\Gamma$ is relevant. Transitions exist for any $\Gamma \neq 0$, with transition temperatures reduced logarithmically for small Γ.

6 Two exactly decoupled limits of the C_3 model

With $\psi = 3\alpha$ the same Hamiltonian reads

$$H = -\sum_{\langle ab\rangle} \big[J_1 \sigma_a\sigma_b + (J_2\sigma_a\sigma_b + J_4)\cos(\psi_a - \psi_b)\big], \tag{11}$$

$J_1 = \frac{160}{81}G$, $J_2 = \frac{128}{81}G$, $J_4 = \frac{512}{729}H_4$, $r \equiv J_4/J_2 = \frac{4}{9} zH_4/zG$.

- $J_4 = 0$ ($H_4 = 0$): σ and $\varphi = \psi + \pi(1-\sigma)/2$ decouple. The octupole sign (TR odd) orders first, at $T_I = 2.269 J_1$; the full octupole φ follows at $T_{\rm KT} = 0.893 J_2$.
- $J_2 \to 0$ ($G \to 0$): σ and ψ decouple. The TR-even hexadecapole ψ ($l = 4$, $m = \pm 3$) orders by itself.

The two limits have opposite order of transitions, so the Ising and KT lines must cross at some $r_\times$. Leading-order estimates of both lines:

$$T_{\rm KT} \simeq 0.893\,(J_2 + J_4\langle\sigma\sigma'\rangle) \xrightarrow{\ \sigma\ \text{ordered}\ } 0.893(1+r)J_2, \tag{12}$$

$$T_I = 2.269\Big[J_1 + \frac{T_I}{2}\ln\frac{I_0\big((J_2+J_4)/T_I\big)}{I_0\big(|J_2-J_4|/T_I\big)}\Big] \quad \text{(tree integration of } \varphi\text{)}, \tag{13}$$

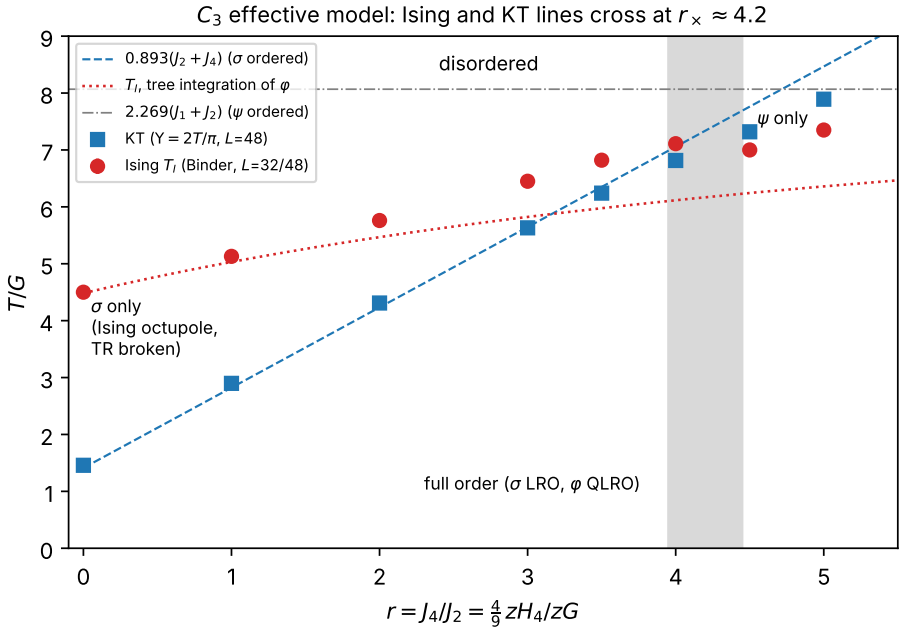
and $T_I \le 2.269(J_1 + J_2) \approx 8.07\, G$ once ψ is ordered.

7 Monte Carlo of the C_3 effective model

Square lattice, $L = 16, 32, 48$, Metropolis plus Wolff clusters in both sectors (σ at fixed φ with bond $J_1 + J_4\cos\Delta\varphi$; embedded φ at fixed σ with bond $J_2 + J_4\sigma\sigma'$; both valid for either sign), annealed downward in T. Ising T_I from Binder crossings and the collapse of $|m_\sigma|$; KT from $\Upsilon_\varphi = 2T/\pi$ at $L = 48$ (twisting φ or ψ gives the same Υ).

r	T_I/G	$T_{\mathrm{KT}}(L=48)/G$
0	4.50	1.46
1	5.13	2.90
2	5.76	4.31
3	6.45(12)	5.63
3.5	6.82	6.24
4	7.11	6.82
4.5	7.00(10)	7.32
5	7.35	7.89

- $r = 0$ reproduces the decoupled values ($2.269J_1 = 4.48$; $0.893J_2 = 1.41$ with the usual $L = 48$ upward shift).
- For $r \leq 2$ the KT line follows $0.893(1 + r)J_2$ and T_I follows the tree estimate.
- The lines cross at $r_\times \approx 4.2 \pm 0.2$, i.e. $zH_4/zG \approx 9.5$. Above it the order is reversed: an intermediate phase with quasi-long-range ψ order and disordered σ appears. This is the TR-even hexadecapolar (vestigial) phase of the tilted tetrahedra; its elementary vortex is a $2\pi/3$ rotation of the body.
- Near $r_\times$ the Binder cumulant occasionally becomes negative at $L = 48$, which may signal a weakly first-order merged transition; the present sampling (no parallel tempering) does not decide this.



8 Dependency chain and next step

$\kappa_l \to (\lambda_E, \lambda_{T_2}) \to \Gamma \to$ selected axis $\to$ effective 2D model with couplings fixed by $(G, H_4) \to$ phase sequence

Open: nature of the transitions near $r_\times$ (parallel tempering, energy histograms, $L \geq 64$) and whether the physical (K, zG, zH_4) window for C_3 selection reaches $r_\times$.